DUTE_AI readings

Site:	UCL Moodle	Printed by:	Yildiz Uzun
Course:	CCME0037: Design and Use of Artificial Intelligence in Education 25/26	Date:	Tuesday, 31 March 2026, 3:24 PM
Book:	DUTE_AI readings		

Description

This book includes:

- Your weekly readings and materials for the module
- Additional supplementary readings and materials which you can consult for your final essay

Note that your readings will either be linked PDFs or will be accessible through the library link.

Table of contents

Week 1: Introduction to DUTE_AI and Learning Analytics

Week 2: Data as the Key Affordance of Technology for Learning

Week 3: Richard Noss Lecture

Week 4: Learning Analytics

Week 5: Multimodal Learning Analytics

Week 6: Value-driven Nature of Data and Algorithms

Week 7: From Prediction to Generation | Natural Language Processing in Education

Week 8: Three conceptualisations of AI and a Vision for Hybrid Intelligence Systems

Week 9: Evidence-informed Implementations of AI in Education

Week 1: Introduction to DUTE_AI and Learning Analytics

Two main books highly relevant to the content of the module that we will refer to frequently:

Wiggins, C., & Jones, M. L. (2023). *How data happened: A history from the age of reason to the age of algorithms*. WW Norton & Company.

Summerfield, C. (2025). These Strange New Minds. Penguin Books.

Core Readings

- *Cukurova, M. (2024). The Interplay of Learning, Analytics, and Artificial Intelligence in Education. British Journal of Educational Technology. https://bera- journals.onlinelibrary.wiley.com/doi/epdf/10.1111/bjet.13514*
- Wise, A. F. (2019). Learning analytics: Using data-informed decision-making to improve teaching and learning. *Contemporary technologies in education: Maximizing student engagement, motivation, and learning*, 119-143.

Further reading materials and other media

- Lang, C., Siemens, G., Wise, A., & Gasevic, D. (Eds.). (2022). *Handbook of learning analytics* (p. 23). New York: SOLAR, Society for Learning Analytics and Research.
- O'Neil, C. (2017). *Weapons of math destruction: How big data increases inequality and threatens democracy*. Crown.
- Ferguson, R. (2012).Learning analytics: drivers, developments and challenges. *International Journal of Technology Enhanced Learning*, *4*(5-6), 304-317.
- Pardo, A., Jovanovic, J., Dawson, S., Gašević, D., & Mirriahi, N. (2019). Using learning analytics to scale the provision of personalised feedback. *British Journal of Educational Technology*, *50*(1), 128-138.

Week 2: Data as the Key Affordance of Technology for Learning

Core Reading

- **Fischer, C., Pardos, Z. A., Baker, R. S., Williams, J. J., Smyth, P., Yu, R., ... & Warschauer, M. (2020).** Mining big data in education: Affordances and challenges. *Review of Research in Education*, *44*(1), 130-160.
- Weidlich, J., Gasevic, D., Drachsler, H., & Kirschner, P. (2025). ChatGPT in education: An effect in search of a cause. *PsyArXiv Preprints*.

Further reading materials and other media

- Suleyman, M., & Bhaskar, M. (2023). The coming wave. Penguin Books. (chapter 2). Here is free access in case you can't find the book as it is new.
- Suleyman, M., & Bhaskar, M. (2023). The coming wave. Penguin Books. (the rest of the book)
- Kevin Kelly (2010). What technology Wants. Penguin Books.
- Chauhan, S. (2017). A meta-analysis of the impact of technology on learning effectiveness of elementary students. *Computers & Education, 105*, 14-30.
- Karpinski, A. C., Kirschner, P. A., Ozer, I., Mellott, J. A., & Ochwo, P. (2013). An exploration of social networking site use, multitasking, and academic performance among United States and European university students. *Computers in Human Behavior, 29*(3), 1182-1192.
- Reich, J. (2020). MIT Media Press.


failuretodisrupt.com
FAILURE TO
DISRUPT

Week 3: Richard Noss Lecture

Core Reading

- Sinha, T., & Kapur, M. (2021). When Problem Solving Followed by Instruction Works: Evidence for Productive Failure. *Review of Educational Research*, *91*(5), 761–798. https://doi.org/10.3102/00346543211019105

Further Reading

- Kapur, M. (2024). Productive Failure: Unlocking Deeper Learning Through the Science of Failing. Wiley Publishing

Week 4: Learning Analytics

Core Reading

- Gašević, D., Dawson, S., & Siemens, G. (2015). Let's not forget: Learning analytics are about learning. *TechTrends*, *59*, 64-71.
- Mangaroska, K., & Giannakos, M. (2018). Learning analytics for learning design: A systematic literature review of analytics-driven design to enhance learning. *IEEE Transactions on Learning Technologies*, *12*(4), 516-534. Please note that the appendices can be found here: Appendix A , B , C , D .

Further Readings

- Kirkwood, A., & Price, L. (2014). Technology-enhanced learning and teaching in higher education: what is 'enhanced' and how do we know? A critical literature review. *Learning, media and technology*, *39*(1), 6-36.
- Dede, C. (2008). Theoretical perspectives influencing the use of information technology in teaching and learning. *International handbook of information technology in primary and secondary education*, 43-62.
- Abrahamson, D., Shayan, S., Bakker, A., & Van Der Schaaf, M. (2016). Eye-tracking Piaget: Capturing the emergence of attentional anchors in the coordination of proportional motor action. *Human Development*, *58*(4-5), 218-244.
- A short summary for Instructional Design Theories and Model



Week 5: Multimodal Learning Analytics

Core reading

- Cukurova, M., Giannakos, M., & Martinez-Maldonado, R. (2020). The promise and challenges of multimodal learning analytics.
- Giannakos, M., & Cukurova, M. (2023). The role of learning theory in multimodal learning analytics. *British Journal of Educational Technology*, *54*(5), 1246-1267.

Further reading

- Spikol, D., Ruffaldi, E., Dabisias, G., & Cukurova, M. (2018). Supervised machine learning in multimodal learning analytics for estimating success in project-based learning. *Journal of Computer Assisted Learning*, *34*(4), 366-377.
- Alwahaby, H., Cukurova, M., Papamitsiou, Z., & Giannakos, M. (2021). The evidence of impact and ethical considerations of Multimodal Learning Analytics: A Systematic Literature Review. https://doi.org/10.35542/osf.io/sd23y
- Alwahaby, H., & Cukurova, M. (2024). Navigating the ethical landscape of multimodal learning analytics: A guiding framework. In *Ethics in Online AI-based Systems* (pp. 25-53). Academic Press.
- Giannakos, M. et al., 2022, The Handbook of Multimodal Learning Analytics, Springer.
- Vinciarelli, A., Pantic, M., Heylen, D., Pelachaud, C., Poggi, I., D'Errico, F., & Schroeder, M. (2012). Bridging the gap between social animal and unsocial machine: A survey of social signal processing. *IEEE Transactions on Affective Computing*, *3*(1), 69-87.

In this episode, Dr Rogers Kaliisa discusses the topic of MMLA with Prof. Mutlu Cukurova from the University College London and Dr Sambit Praharaj from the Center for Advanced Internet Studies (CAIS) in Germany. Listen to this episode on Soundcloud, iTunes and Spotify.

Week 6: Value-driven Nature of Data and Algorithms

Core reading:

- **Winner, L. (1986).** The Whale and the Reactor: A Search for Limits in an Age of High Technology. (The entire book is relevant but particularly read chapter 1.2 - Do artefacts have politics?)
- Baker, R.S., Hawn, A. (2022) Algorithmic Bias in Education. International Journal of Artificial Intelligence in Education 32, 1052–1092. https://doi.org/10.1007/s40593-021-00285-9

Further Reading:

- Edwards, P. (2010). A Vast Machine: Computer Models, Climate Data, and the Politics of Global Warming. MIT Press.
- O'Neil, C. (2016) **Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy** . New York, NY, Broadway Books.
- Bowker, G., & Star, S. (2000) Sorting Things Out: Classification and Its Consequences. MIT Press
- Hattie, J., & Timperley, H. (2007). The power of feedback. Review of educational research, 77(1), 81-112.
- Van Merriënboer, J. J., & Kirschner, P. A. (2017). Ten steps to complex learning: A systematic approach to four-component instructional design. Routledge.
- Luckin, R. (2008) . The learner-centric ecology of resources: A framework for using technology to scaffold learning. Computers & Education, 50(2), 449-462.
- Knobbout, J., & van der Stappen, E. (2018).Where Is the Learning in Learning Analytics?. In European Conference on Technology Enhanced Learning (pp. 88-100). Springer, Cham.
- Martinez-Maldonado, R., Gaševic, D., Echeverria, V., Fernandez Nieto, G., Swiecki, Z., & Buckingham Shum, S. (2021). What Do You Mean by Collaboration Analytics? A Conceptual Model. *Journal of Learning Analytics, 8*(1), 126-153.

Week 7: From Prediction to Generation | Natural Language Processing in Education

Core Reading

- Giannakos, M., Azevedo, R., Brusilovsky, P., Cukurova, M., Dimitriadis, Y., Hernandez-Leo, D., ... & Rienties, B. (2024). The promise and challenges of generative AI in education. *Behaviour & Information Technology*, 1-27.

Further Reading

- Pashler, H., Bain, P. M., Bottge, B. A., Graesser, A., Koedinger, K., McDaniel, M., & Metcalfe, J. (2007). Organizing Instruction and Study to Improve Student Learning. IES Practice Guide. NCER 2007-2004. *National Center for Education Research*. (Look at Table 2 specifically, can you model AI to deliver those principles?)
- Cukurova, M., Khan-Galaria, M., Millán, E., & Luckin, R. (2022). A learning analytics approach to monitoring the quality of online one-to-one tutoring. *Journal of Learning Analytics*, *9*(2), 105-120. (Look at table 1 specifically, can you model AI to deliver those principles?)
- Miao, F., & Cukurova, M. (2024). AI Competency Framework for Teachers. UNESCO Publishing.
- Luckin, R., & Cukurova, M. (2019). Designing educational technologies in the age of AI: A learning sciences-driven approach. *British Journal of Educational Technology*, *50*(6), 2824-2838.
- Daniel, K. (2017). Thinking, fast and slow.
- Wang, R. E., Ribeiro, A. T., Robinson, C. D., Loeb, S., & Demszky, D. (2024). Tutor copilot: A human-ai approach for scaling real-time expertise. arXiv preprint arXiv:2410.03017.

Week 8: Three conceptualisations of AI and a Vision for Hybrid Intelligence Systems

Core reading:

- Molenaar, I. (2022). Towards hybrid human-AI learning technologies. *European Journal of Education*, *57*(4), 632-645.
- Järvelä, S., Nguyen, A., & Hadwin, A. (2023). Human and artificial intelligence collaboration for socially shared regulation in learning. *British Journal of Educational Technology*, *54*(5), 1057-1076.
- Cukurova, M., Suraworachet, W., Zhou, Q., & Bulathwela, S. (2025). *Towards Synergistic Teacher-AI Interactions with Generative Artificial Intelligence*. arXiv. https://doi.org/10.48550/arXiv.2511.19580

Further Reading:

- Cukurova, M. (2019). Learning analytics as AI extenders in education: Multimodal machine learning versus multimodal learning analytics. In *Artificial intelligence and adaptive education* (Vol. 2019). AIAED.
- Bennett, M. (2023). *A brief history of intelligence: evolution, AI, and the five breakthroughs that made our brains*. HarperCollins.

Week 9: Evidence-informed Implementations of AI in Education

Core Reading

- Wrigley, T. (2018). The power of 'evidence': Reliable science or a set of blunt tools?. British Educational Research Journal, 44(3), 359-376.
- Du, Y., Borchers, C., & Cukurova, M. (2025). Benchmarking Educational LLMs with Analytics: A Case Study on Gender Bias in Feedback. *arXiv preprint arXiv:2511.08225*.

Further Reading

- Borges, B., Foroutan, N., Bayazit, D., Sotnikova, A., Montariol, S., Nazaretzky, T., ... & Bosselut, A. (2024). Could ChatGPT get an Engineering Degree? Evaluating Higher Education Vulnerability to AI Assistants. *arXiv preprint arXiv:2408.11841*. (See the appendix for all the prompting strategies)

- Cukurova, M., & Luckin, R. (2018). Measuring the impact of emerging technologies in education: A pragmatic approach. Second Handbook of Information Technology in Primary and Secondary Education, 1-19.
- Cukurova, M., Luckin, R., & Clark-Wilson, A. (2019). Creating the golden triangle of evidence-informed education technology with EDUCATE. British Journal of Educational Technology, 50(2), 490-504.
- Reeves, T. C. (2008). Evaluation of the design and development of IT tools in education. In J. Voogt, & G. Knezek (Eds.), International handbook of information technology in primary and secondary education. Berlin/Heidelberg/New York: Springer.
- Simpson, A. (2017). 'The misdirection of public policy: Comparing and combining standardised effect sizes', Journal of Education Policy, 32(4): 450–466.
- Cukurova, M., Kralj, L., Hertz, B., & Saltidou, E. (2024). Professional Development for Teachers in the Age of AI. European SchoolNET.